Multi-Disciplinary University-Industry Interaction – Case Mikkeli

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Abstract

There is a challenge in university-industry interaction for finding ways for effective multi-disciplinary cooperation. Multi-disciplinary aspect is vital for companies' future innovation. This is also a challenge for universities once they are finding ways to foster education towards multi-disciplinary processes. The cooperation between multiple disciplines remains an uncharted terrain in the context of university-industry cooperation. This paper presents examples of two university-industry cooperation assignments conducted multi-disciplinarily. The article is introducing the views of teachers, students and company representatives involved in the assignments. The objective of this research is to find out how facilitation should be conducted for effective implementation.

The commissioning companies of the study are Finnish industrial companies searching for new business opportunities. The assignments are focusing on finding answers for specified problems the companies are facing. The students participating in these assignments are representing multiple degree programmes (business management, environmental engineering, and material technology). Students bring their own discipline related competences to the projects. This combination enables the new solutions for the problem that companies may use to develop their business.

The research method used was interviews and focus group discussions. Interviews were conducted with company representatives whereas focus groups were used to find out insights of the teachers and students involved. Research methods are used to find good practices during the processes and to determine what obstacles lie in the way of cooperation.

As a result, good practices and the developmental needs are identified. The results propose the practices to be taken in the future assignments. These results can be taken into consideration once future practical implications are planned. This applies both to the universities and to the companies.

Multi-disciplinary projects are strongly needed in contemporary business to find new business opportunities. The cooperation between different parties needs good practices as template on which new projects can be established. Article lends a hand on that need. However, the implications proposed must be verified in real life assignments. This can be focus of future studies.

Keywords

Multi-disciplinary cooperation, product development, university-industry interaction, university-industry cooperation, university-industry collaboration

1 Introduction

The Finnish universities have multiple goals to achieve. Two of them are examined within this paper. Firstly, the universities have to deliver skills of innovation management to students. Secondly, there is a need for cooperation with local companies. This paper examines how these two goals could be brought together.

For this report there are two university-industry cooperation assignments studied. Both assignments are commissioning works for local companies. Assignments are carried out by the students studying in Mikkeli University of Applied Sciences. Students are representing multiple disciplines. The formation of student teams is executed multi-disciplinarily. In this manner, it is expected to find out better and more novel solutions for assignment problems involved.

To make assignments productive there are multiple factors that have to be considered. This paper is trying to frame what kind of expectations and needs there exist in the insights of teachers, students and company representatives taking part in the assignments. The objective of this research is to find out how assignment facilitation should be conducted to secure the implementation of assignments.

The participants for the two company assignments were interviewed to gather their insights about the assignments. Company representatives and teachers were interviewed individually. Focus group meeting was arranged for students to find out their thoughts about the assignments. Based on these results a content analysis is conducted displaying the good practices needed for productive assignments implementation. This can be used as a tool for planning multi-disciplinary company cases in the future.

The paper begins with literature review of innovations, university-industry interaction and multi-disciplinary activities. In the second part, the methods used are explained. The paper unfolds with the results drawn from data. Based on these results there are good practices and a framework of productive future assignments displayed. In the end there is discussion focusing on limitations, recommendations and suggestions for further research.

2 Literature review

In this chapter, the key concepts used in the paper are presented. First, the concept of innovation is explained. Second, the university-industry interaction and its possibilities are covered. Third, the multi-disciplinary approach for university-interaction is clarified.

2.1 Innovation

The concept of innovation, as understood in this paper, has its origins in the definition of Schumpeter (2013). According to this definition there has to be continuous process of innovation that is producing innovations for enterprises to survive. According to Benner and Tushman (2003) innovations may occur in four different ways. Benner and Tushman illustrate the development of innovations in a two-dimensional model. These dimensions are

(1) a new innovation in relation to the existing technology

(2) innovation's relation to the existing segment of the market.

In the extreme of technological dimension, there is an incremental innovation, which produces some improvement for existing solution. At the other extreme, there is a structural innovation, which often requires exceeding the current technological standards of the organization's current knowledge. On customer relationship innovation dimension, the extremes are the product offered for current customer segment and new product launch for new customer segments. All the existing and novel innovations can be positioned within this framework of innovations.

To enable innovations, organisations must arrange the environment to promote desired activity. This is made possible in adhocracy organisation. Adhocracy is an organization that is horizontally very specialized, but vertically flat. The informal organization culture allows a flexible environment for rapid response requirements. (Robbins 1990.) What is characteristic for adhocracy is a gathering of experts from different fields to work on same subject or problem. Adhocracy is typical for modern project organizations. The dynamic way of working allows a broad participation for the various members of the project team in their own areas of expertise. The key challenge in project organizations is the poor transformation of information into the form of organization's explicit knowledge (Nonaka and Takeuchi 1995). This must also be noticed in project team collaboration.

2.2 University-industry interaction

The demand for university-industry cooperation is widely recognized to foster innovations (e.g. Mansfield 1998; Meyer-Krahmer and Schmoch 1998; Etzkowitz 1998; Ambos et al. 2008). This field of research is studied from multiple perspectives. Etzkowitz (1998) presents the need of university-industry cooperation. This concept is referring to the mutual collaboration, which is aiding both parties, university and enterprise, achieving their own goals. On university side, this is generally seen as support for research agendas, whereas the industry gains innovations and support for service and product development. These actions are seen as a vital part of technology transfer happening in three forms. These approaches can be characterized as

(1) hands off - leaving the matter entirely to the transfer office;

(2) knowledgeable participant being aware of the potential commercial value of research and willing to play a significant role in arranging its transfer to industry;

(3) and seamless web that integrates campus research group and research program of a firm.

The same perspective is also found in Lee's (1996) research, which is focusing only on the cooperation between staff members of the university and enterprise. Both of these researches are forgetting the input of students involved in the projects carried out.

Bruneel et al. (2010) have set the focus of their research on the factors that diminish the barriers to university-industry collaboration. Their results state that prior experience of

collaborative research lowers orientation related barriers and that greater levels of trust reduce barriers. Findings also indicate that breadth of interaction diminishes the orientation-related, but increases transaction-related barriers. (Bruneel et al. 2010.) Therefore, it can be assumed that the relationship between university and industry must be established somewhat earlier, before the student projects can be started.

The university-industry interaction research is also focusing on the channels used to deliver information between universities and industry. Bekkers and Freitas (2008) have conducted this kind of research. They have studied which channels' importance is seen the most vital in the cooperation of universities and industry. In this research, the importance of students working as trainees is seen quite high. This channel's average was 2.8 on a scale 1-5, where 5 is the most significant grade. On top of this list are scientific publications. Industry representatives gave it a grade 3.1 whereas university representatives gave grade 3.5. According to this results less important channels are e.g. contract research (excl. Ph.D. projects), flow of university staff members to industry positions (excl. Ph.D. graduates), consultancy by university staff members and contract-based in-business education and training delivered by universities. This is interesting especially because D'Este and Patel (2007) have found those to be the most used channels between universities and industry from the point of view of the academics. Based on these notions it is seen as quite important that the students are participating in the company activities along with the academics.

Even though the top universities are the most wanted partners for collaboration, local educational partners are also needed. Mansfield and Lee (1996) have been researching the collaboration between university and industry. They found out that companies taken part into collaboration considered the academic research findings as most valuable output in product and process development. This means that in product development orientation the proximity to industry might play a key role once choosing partners to collaborate with. This applies to both the university and industry. Freitas et al. (2013) propose that university research and development projects with firms in emergent industries are less likely than projects with firms in mature industries to be the result of academic initiatives. Therefore, the contacts towards industrial companies should be targeted to the enterprise segments following these criteria.

The cooperation of universities and industry may include participation of third parties as well. The most common third party is governmental actors presented by triple-helix model (e.g. Etzkowitz and Leydesdorff 2000; Leydesdorff 1995). This model relies on the idea of involving more parties into the innovation process to increase the occurrence of very new ideas. This can be seen as a preform of open innovation (Chesbrough 2006 or Perkkmann and Walsh 2007) relying on multiple partners with different backgrounds as contributors to innovations. The third party involved can be understood as students, as it is done in this paper.

2.3 Multi-discipline activities

The concept of multi-discipline activities is sometimes used as synonym for inter-disciplinary and cross-disciplinary activity. The focus of academic research has been on the role of multi-disciplinary teamwork during the course of knowledge creation (West and Anderson 1996; McFadyen and Cannella 2004; Taylor and Greve 2006; Chang 2008). The reason for using such teams is that when representatives from all of the relevant disciplines are brought together, team decisions and actions are more likely to encompass the full range of perspectives and issues that might affect the success of a collective venture (Van Der Vegt and Bunderson, 2005).

Prohl (1997) has made a literature review of 26 articles published between 1992 and 1995, and noted that many of the definitions for what constitutes a team share the following characteristics that can be thought of as requirements for effective multi-disciplinary cooperation as well:

› *Shared identity*. Meaning that all members of a team perceive the team as a functional unit and perceive themselves as part of that unit.

› *Common goal*. Meaning that all members of the group are working towards accomplishing the same task.

› *Interdependence*. Meaning that the efforts of all group members are necessary to complete the task.

› *Personal interaction*. Meaning that the members of the group communicate with each other and share information.

› *Mutual influence*. Meaning that each member of the group influences every other member's understanding of the task and their means for accomplishing the task.

Prohl (1997) noted that organizations might form teams to accomplish any number of tasks such as to develop new products, to re-engineer processes, to improve customer relations or to improve organizational performance. The level of expertise does not necessarily have to be superior for the team to excel. Instead Ma et al. (2014) have found out that moderate level expertise knowledge disparity produces novel combinations and higher-level experience knowledge variety facilitates exchange, which increases the quality of knowledge creation. That is especially the benefit that multidisciplinary assignments are relying on.

2.4 Multi-discipline action in university-industry interaction

The multi-discipline teams must be brought to the field of university-industry interaction. The role of students in the collaboration of universities and industry seems to author's knowledge an uncharted terrain. These kind of activities need new practices to develop novel ways of creating collaboration. The multi-discipline teams might be a proper way to conduct university-industry interaction and involve students in the activities.

As a result, it is expected that novel ideas are generated as proposed by Van Der Vegt and Bunderson (2005). Multi-disciplinary teams can create technologically novel innovations or find new market segments for existing solutions as proposed by Benner and Tushman (2003). The multi-disciplinary approach is also following the ideal of adhocracy presented by Robbins (1990).

3 Methodology

There are two research methods used in this paper: focus group meeting and semi-structured interview. Focus group is used to find out insights, needs and expectations of students that participated in the assignments. Semi-structured interview is used as research method to interview the company representatives and university personnel.

Focus group is a method that is used to research opinions of several individuals in one specific situation simultaneously. Eriksson and Kovalainen (2008) present the focus group as a joint meeting, gathered by researchers. The participants are selected based on the experience that they have on the topic that is subject of the research. In this research, the experience on the commissioning company assignment is the reason to choose the students to participate in the focus group meetings. The benefit of focus groups is based on the interaction of the participants. Once the participants are sharing their information about the topic amongst other participants, it can encourage the other participants to share their own experiences. As a result, discussions arise that respondents could not independently create without the help of the rest of the participants. In many cases, the experiences of other participants help other members to remember and recall their own experiences or set them in new context once they have heard how the other participants have experienced the same kind of circumstances. (Puchta and Potter 2006.) This enriched information is one of the benefits why the focus group is selected as research method. In this research one focus group meeting was arranged. There were in total four students attending from the total number of 10 students that participated in the two assignments. These results are used to represent the students‘ view on the assignments.

The interviews of the staff members are conducted as individual interviews. Interviews were executed using semi-structured interview as research method. One company representative was interviewed from each of the company assignment enterprises. These interviews are used as representations of the companies‘ needs and expectations. The commissioning companies do not have any connection to each other. Therefore, it is expected that bringing them together isn’t going to offer additional information.

Along with the company, representative interviews there are three interviews executed for the university staff members: two teachers and one business liaison officer. This part of the research material is representing the needs and expectations of the university.

Focus group meeting and interviews covered all the parties that took part in the assignments. All the parties are represented with more than one representative. The

assignments the participants are representing are two assignments that were conducted during academic year in Mikkeli University of Applied Sciences. In this manner, the data is showing holistic needs of the participants. Research must yet be considered as intensive case study trying to understand the features of these particular cases.

The commissioning companies of the study are Finnish industrial companies searching for new business opportunities. The assignments are focusing on finding answers for specified problems the companies are facing. The students participating in these assignments are representing multiple degree programmes (business management, environmental engineering, and material technology). The details of the participants are not revealed to ensure the anonymity of the participants. The focus group meeting and interviews were recorded and transcribed. The analysis of the data was done using content analysis. The needs and expectations of the participants explored from the data using open coding technique. The framework for effective multi-disciplinary company cases was constructed based on these findings.

4 Results

In this chapter the results are presented in three parts. First, the results from the focus group meeting are displayed. Second, the interviews of the university representatives are brought together. Third, the answers from the company representative interviews are presented.

4.1 Students' perspective

In general, the students thought that the multi-disciplinary approach is a good idea. It is also good to have real companies involved as part of their studies. The cooperation between different degree programmes did not work out easily. There was a slight difference in the working cultures of degree programmes. This was seen as somewhat challenging in the cooperation. The roles and responsibilities were not clear enough causing some confusion about who should be working with each fields of the assignments. The cooperation between different disciplines needs improving. It did happen in these cases, but the students saw that there might be more joint activities.

The assignments that the students were working on were seen as quite good ones. Those were not too loose or too tight. It was pointed out that there should be some connection with the assignment and students' background. Otherwise, the students are not able to give their full contribution for the project.

Assignments were connected to the syllabus of the degree programme. The students saw that the availability of study material should be considered. Sometimes students also found it hard to determine if they were heading in the right direction. The need for virtual communication among students was also discussed. Once the students are from different locations it does set a need for virtual communication.

The motivation to take part in company activities is not taken for granted. For fluent cooperation it was seen that motivation should be checked already in the recruiting phase with letter of motivation. During the process, there should be compulsory meetings that have to be attended. The ones not participating should be expelled from the project.

A project manager is needed to carry out assignments more effectively. It was noted that the project manager should be a staff member. The project plan with pre-scheduled meeting was also seen as a tool to make the process more straightforward.

The schedule of the assignments was fine. However, different deadlines should be set in the very beginning of the project. If there is too much idling time, it is not really helping the assignment results.

The visiting opportunity to the commissioning company was seen as an important feature of the assignments. A deeper understanding of the company processes was one of the key things that students did expect from the assignment. Communication and cooperation with companies was working properly during process. Although, sometimes there were a bit longer delays before emails were responded. Students also remarked that the way in which the company does its own part is affecting the results that the students are able to achieve.

4.2 University's perspective

There were different roles that the interviewees played in the process. One gathered the assignments from the companies whereas two worked as teachers during the assignments. It was seen that teachers' role should be strengthened during the process. The responsibilities of the different staff members should also be clarified. It was proposed that there should be at least one multi-disciplinary assignment contact person in each discipline. The role of the teacher was considered more as a mentor than a traditional instructor.

The process was seen a bit unclear. The students' progress was not clear enough for staff members. Therefore, it was proposed that there should be some deadlines during the process that would ensure the progress of assignments in the right direction. Student generated project plan and meetings were mentioned as possible tools to achieve this goal.

The schedule of the processes was considered satisfactory in general. The developmental need was seen with the exact deadline dates that should be confirmed in the very beginning of the process.

The cooperation with the companies was considered unproblematic. During the process, the students were doing most of the communication with the company representatives. It was noted as very important that the company representatives are devoted to the case and that there is a possibility to visit the company during the process.

The reporting of the results was done following the needs of commissioning cases. The results were shown either in presentation or in written document. This was considered as a good approach. However, there remains a need for deeper understanding of the students' learning process.

The cooperation between disciplines was not seen as completely functioning. There should be more reasoning for the students outlining the possibilities of the multi-disciplinary activities. Otherwise, the different approaches are not brought together effectively.

The main limitation for one's own work was the lack of time available to guide assignment. Part of the interviewees were not familiar with multi-disciplinary actions before, whereas some had previous experience. Because of the novelty of the multi-disciplinary concept it was considered as vital to spread practices and experiences for staff members to encourage the participation in future assignments.

To ensure the continuity, it is important that there are a sufficient number of students involved in the assignments. This requires proper marketing for students and staff members. The student recruiting does have to consider student requirements. It was stated that the students participating should be highly social and capable of cooperating with different kind of people. It was also noted that the students must have some kind of connection to the assignment to ensure the commitment to the project.

4.3 Companies' perspective

The companies were very contented with the assignments. The results of the assignments were at some level seen as surprisingly good.

The communication during the process was unclear from time to time. It was not always certain who should be involved in communication in different occasions. Email was considered as a good tool for communication, because it does always leave a trail. Along with this more meetings during process would have been needed. In current situation there were only the launch and presentation of the results. It was also thought that prior knowledge of the students' background would increase the understanding of the students' capabilities.

The form of report was depending on the needs of the company. There were needs for presentation form and report form. Presentation was seen as a good tool to be used within company once presenting assignment results for different stakeholders. The report was useful as used in future project papers. It was also seen important that the possible tests during the process would be documented along with the final results.

The process itself was thought as satisfactory. The number of project meetings could be increased. It would be possible to work in company premises sometimes. The participation of the university staff members was seen as upright.

The schedule of the assignments was considered as tight. It would have been possible to take even more time for the assignments. What must be decided are the deadlines and

milestones for the project in the beginning of the project. The amount of time taken from the company representatives was seen as reasonable. It was also noted that if the company is expecting any kind of results from the assignments, there needs to be serious involvement from that party as well.

There was also a note on the assignment itself. If the company truly needs the results of the assignment, it is more beneficial for the company and for students. Therefore, the company should pay attention to the generation of the assignment and its details. In this manner, the results will be much better.

Non-disclosure agreements were also considered as a very important part of the assignments. As the students are working with very crucial business information, high confidentiality is expected during the assignment. This aspect must be emphasized because it is a very different situation compared to normal studies.

5 Discussion

Students, university staff members and company representatives have many different needs for the cooperation. Some of them are common for all parties, but there are also needs that differ. All of them must be considered once planning multi-disciplinary assignments in future. In the following table, the views of students, staff members and company representatives are brought together. On top of the table are the common views, whereas the ones shown only on one party's view are below. The order of the objects in the table is not prioritized.

	Students	**Staff members**	**Company representatives**
Common needs	Schedule and deadline		
	Joint meetings		
	Devotion to assignment		
Differentiated	Clarified student roles	Clarified teacher roles	More information of students backgrounds
	Project plan and manager	Project plan	Reporting the process
	Connection between assignment and student background	Form of report based on the needs of the company	Form of report based on the needs of the company
	Tools for virtual communication	Responsible teacher for each discipline	Clear communication during process

ee ds	Common goal for team members	Justification of multi-disciplinary activity	NDAs
	Student motivation verification	Good practices visible	
		Marketing for students and staff members	
		Student requirements	

Tab. 1: Needs of parties involved in assignments.

Common for all parties is the need for scheduling the project at the beginning. The project plan should include meetings that are held regularly during the process. In between the meetings the students are capable of working on their own, as they are aware of their own roles. All the participants must be devoted to the assignment to ensure valuable results.

The roles of different parties are also important. The companies do know much better, what they can expect from assignments as they have better understanding of their backgrounds. This helps to adjust the assignments to the capabilities of the students. The prior planning of the project as well as the visibility of the process during work was also considered important. The reporting must be done in a form that helps the commissioning company to achieve its goals. The communication during the process might be done using virtual tools. This enables flexible cooperation between different parties.

Along with these aspects, the common goal must be clearer. Adhocracy (Robbins 1990) and its superior features must be further justified for students. The requirements of shared identity, common goal, interdependence, personal interaction and mutual influence (Prohl 1997) did not happen in all levels. These do require that there is enough effort put on student assignment recruitment. The requirements of the students must be considered during the recruitment. To ensure this there needs to be a responsible teacher in each discipline who is helping in student recruiting. The teachers need at least basic information about the multi-disciplinary approach in form of good practices. To ensure the confidentiality, non-disclosure agreements (NDAs) must be made with all the students participating in the assignment.

As these aspects are covered, the assignments in general can be carried out effectively.

6 Conclusions and recommendations

This paper has been examining two assignments conducted in Mikkeli University of Applied Sciences. Assignments are carried out multi-disciplinarily. The needs and expectations of students, staff members and company representatives are researched

using focus group meetings and interviews as research methods. Based on the findings there suggestions about how the assignments should be carried out are presented.

The paper's approach to the topic is very pragmatic. The assignments are showing only a glimpse of emerging phenomenon. Therefore, the results of the research should not be generalized on any level. However, the reforming of university field in Finland does require new forms of interaction between universities and industry. In this approach, new methods and models are needed to be examined. For this purpose, this paper does help the field by giving information about the different needs of the parties.

This paper is presenting a holistic view of the different parties involved. In this manner, it is giving insight on how the different needs interact with each other. There seems to be no conflict between the needs of different parties. However, there do remain parts of the cooperation that must be developed further. These identified needs act as proposals for future research. The future research could be focusing on the framework that is constructed in this paper. It can act as a basis on which the future assignments are constructed. The future research can be focusing on the satisfaction with the improved model among students, university staff and company representatives.

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